

Diagnostics

Choose commands in the Diagnostics menu to access diagnostics. Diagnostics included in the C-200MT program ensure cutter performance. Diagnostics may need to run when mechanical parts have been replaced or new software has been loaded.

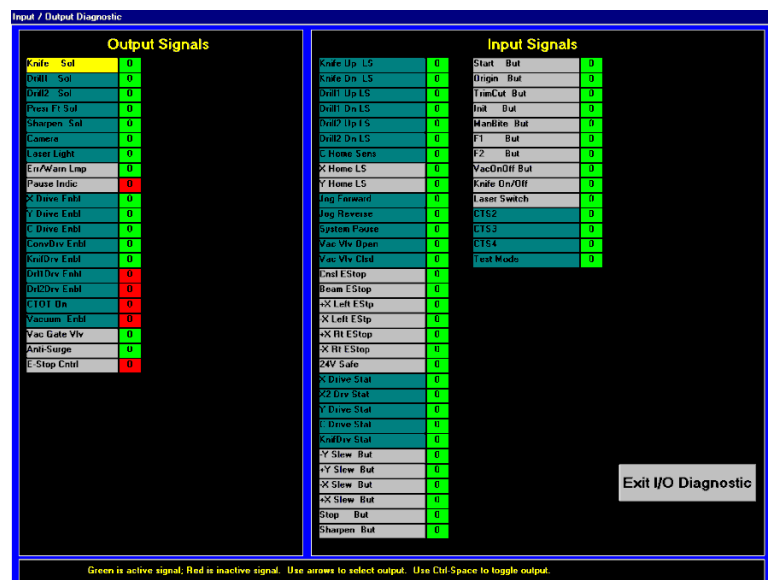
CAUTION: Diagnostics should be used *only* by the person in your company who has received GERBERcutter maintenance training or by Field Service.

Input Output Test (DG 23)

This diagnostic can be used to test individual motors, switches, lights, and other components. Only use this diagnostic when told to by Field Service or in the *C-200MT Software User's Guide* or GERBERcutter operator or system service manual (for example, to diagnose MCC error messages).

To run the diagnostic

1. Turn on the GERBERcutter and run the C-200MT program.
2. Optional — Enable servo power and initialize the cutter.
3. From the Diagnostics menu, select **Input Output Test (DG 23)** to open the DG 23 screen (example shown below).



WARNING!



Stand away from moving parts on your cutter when you use this diagnostic. Injury or death may result if you are in the path of cutter parts when they move. Depending on which control you are testing, part of the cutter may move. This can be, but is not limited to, the knife blade, drill, conveyor, and other mechanical parts.

4. To test an output signal, use the arrow keys or pointing device to highlight the signal that you want to test. Hold down the Ctrl key and press the Space bar.

Note: Some GERBERcutters provide a second, loopback bit that reports the actual state of output. When the system is communicating correctly, output command and loopback signal show the same result (for example, **00**).

In the left half of the screen:

- Active output shows as **green**
 - Loopback signal that does not match active output choice shows as **yellow** (see note above).
 - Inactive output shows as **red**
5. To test an input signal, you will need to use a cutter component. For example, press a button.

The listed signal changes either from 0 to 1 or 1 to 0. The actual component on the cutter you are testing becomes active or inactive.

In the right half of the screen:

- Active input shows as **green**
 - Inactive input shows as **red**
6. After you finish testing the components, press **Esc** or click the **Exit I/O Diagnostic** button.

Servo I/O Test (DG 16)

The Servo I/O Test diagnostic can be used to test X, Y, C servo motors and the Knife, Drill, and Conveyor drives. Only use this diagnostic when told to by Field Service.

How It Works

Servo motion results by the MCC board sending command voltage signals to the motor drives and receiving position information from the encoders. In this diagnostic, the MCC can output a pre-selected command voltage to a selected drive depending on whether the EStop loop is:

- OPEN – Full command voltage is available and can be monitored at various test points in the control.
- CLOSED – Limited command voltage is available and can be used to drive the motors at a reduced speed.

In addition to testing the outputs, the diagnostic can read from the conveyor tachometer and X, Y, and C encoders. On some cutters, the KI sensor and knife speed can also be read.

To run the diagnostic

1. Turn on the GERBERcutter and run the C-200MT program.
2. From the Diagnostics menu, select **Servo I/O Test (DG 16)**.
3. The system opens the EStop loop and displays the Servo Diagnostic screen.
4. Use this screen to:
 - Test Outputs
 - Test Inputs
 - Test Encoders
5. When finished testing, press **Esc** or click the **Exit Servo Diagnostics** button.
6. Clear any error message and then enable servo power.

Testing Outputs with Servo I/O Diagnostic

You can test outputs with the EStop loop either OPEN or CLOSED.

To test outputs with EStop loop OPEN

1. In the Output section of the Servo Diagnostic screen, use the Arrow keys or pointing device to select the output you want to test.
2. Type a value for the desired output voltage (valid range is -10.00 to +10.00 volts).
3. Press the **Ctrl** key to apply the voltage.
4. Monitor the output voltage at test points on the transition board or directly on the selected drive board. Refer to the service schematics for your cutter.

To test outputs with EStop loop CLOSED

1. Double-click on the **Toggle E-Stop State** button.
2. When prompted, type the Field Service level password and press **Enter**.
3. Enable servo power.

EStop STATUS is now Closed.

4. Notice the position of the beam and head on the table.

WARNING!



Be aware of the direction the beam and head will move when a command voltage is applied or injury could result. Refer to the chart below.

<u>Device/Polarity</u>	<u>Result</u>
X1 drive / Negative voltage	Beam moves +X
X1 drive / Positive voltage	Beam moves -X
Y drive / Negative voltage	Beam moves +Y
Y drive / Positive voltage	Beam moves -Y
C drive / Negative voltage	C axis rotates CCW
C drive / Positive voltage	C axis rotates CW
X2 drive / Negative voltage	Beam moves -X
X2 drive / Positive voltage	Beam moves +X
Knife drive / Positive voltage	Knife rotates (if knife switch is in ON position)
Conveyor drive/ Negative voltage	Conveyor moves forward
Conveyor drive / Positive voltage	Conveyor moves reverse
Hvy Drill speed / Positive voltage	Drill rotates (if knife switch is in ON position)
CTOT drive / Negative voltage	CTOT moves forward

5. In the Output section of the Servo Diagnostic screen, select the output you want to test.
6. Type a value for the desired output voltage (valid range is -3.00 to +3.00 volts).
7. Press the **Ctrl** key to apply the voltage. As long as this key is held down, the motor will turn.

Testing Inputs with Servo I/O Diagnostic

To test inputs with EStop loop CLOSED

1. Apply a command voltage to the knife and conveyor drives.
2. In the Input section of the Servo Diagnostic screen, monitor the feedback voltage that displays in the corresponding field.
3. To test the Knife Intelligence sensor:
 - Make sure the Knife motor is turned OFF before applying lateral pressure to the lower guides to move the sensor.
 - Monitor the feedback voltage.

Testing Encoders with Servo I/O Diagnostic

To test encoders with EStop loop OPEN

1. Double-click on the **Reset Encoder Counts to 0** button.
2. At the cutter, physically move one of the servo axes or rotate an encoder while monitoring encoder counts in the Encoder section of the Servo Diagnostic screen.

To test encoders with EStop loop CLOSED

1. Double-click on the **Reset Encoder Counts to 0** button.
2. Apply a command voltage to move the conveyor or servo axis.
3. Monitor encoder counts in the Encoder section of the Servo Diagnostic screen.

KI Calibration (DG 20)

This diagnostic is used to calibrate the Knife Intelligence deflection voltage sensor (if the cutter can utilize KI).

Note: To check if calibration is required, first use the Null KI command.

To run the diagnostic

1. Turn on the cutter and run the C-200MT program.
2. Enable servo power.
3. From the Diagnostics menu, select **KI Calibration (DG 20)** to open the DG 20 screen.
4. Use Trim Cut to lower the knife.
5. On the presser foot bowl assembly, loosen the clamping screw that locks the transducer adjusting screw.
6. Adjust the transducer screw until the output voltage is at, or near, zero.
7. Tighten the clamping screw to lock the transducer adjusting screw.
8. When finished, press **Esc** or click the **Exit DG20** button.
9. Use the Null KI command to null Knife Intelligence.

C Axis Angle

Use this diagnostic to check C-axis rotation. A C-axis that is fully functional, both electrically and mechanically, can turn to any heading (angle).

To run the diagnostic

1. Turn on the GERBERcutter and run the C-200MT program.
2. Enable servo power.
3. Initialize the cutter.
4. From the Diagnostics menu, select C **A**xis Angle.
5. In the C Axis Angle Diagnostic dialog box, type the desired angle. Press **Enter** or click the **Apply** button.

The C-axis rotates to the angle you entered.

6. Repeat step 5 to try another C-axis angle.
7. When finished, press **Esc** or click the **Done** button.

Sharpen Angle

The Sharpen Angle diagnostic, which is used on S-3200 and GT 3250 cutters, allows you to:

- Adjust the sharpen angles
- Sharpen one or both sides of the knife

Settings made using this diagnostic can be saved to all system setup files.

To run the diagnostic

1. From the Diagnostics menu, select **S**harpen Angle.
2. In the dialog box, type to enter new angles for sharpening side one or side two.
3. Type to enter the number of times you want to sharpen both sides.
4. Press the appropriate Sharpen button to test your changes.
5. When finished, press **Enter** or click the **OK** button.

6. When prompted, you can choose to save changes to all setups.
7. If you clicked Yes in step 6, the system prompts that x number of setups were changed. Press **Enter** or click the **OK** button.

Calibrate Conveyor Move

This diagnostic can be used to calibrate conveyor movement over small distances in order to improve precision. Use this diagnostic only when C-200MT software is first installed or when conveyor parameters are reset.

When selected, the conveyor begins by moving 20 inches. This initial movement resets the value of the Ramp Down Dist – Auto Bite parameter. Additional small movements (beginning at 0.1 inches, increasing in 0.1 inch increments up to a move of 8.0 inches) are used to create a reference table that is utilized and updated during normal cut processing.

To run the diagnostic

1. Before starting, do **all** of the following:
 - Clear the table surface of material
 - Turn the CTOT off
2. From the Diagnostics menu, select **Calibrate Conveyor Move**.

The Ramp Down Dist – Auto Bite parameter is reset.

WARNING! To prevent injury, stand clear of the beam during this procedure. As needed, the cutting head is repositioned to the take-on end of the table until all movements are completed.

C Home to System Zero

The C Home to System Zero diagnostic allows you to adjust this knife setting and command the system to plunge the knife four times in a straight line. Settings made using this diagnostic can be saved to all system setup files.

Note: The Supervisor or Field Service password must be entered in Parameter Manager before using this diagnostic.

To run the diagnostic

1. Place a large square of paper on the cutting surface. Position the head over the paper.
2. From the Diagnostics menu, select **C Home to System Zero**.
3. Press the Plunge Knife 4 Times button.
4. After the knife has plunged four times, compare the cuts to the example shown in the dialog box. Make any necessary adjustments.
5. In the dialog box, type the number of degrees to adjust the knife setting.
6. Reposition the head over the paper. Repeat steps 3 through 5 until satisfied with results.
7. When finished, press **Enter** or click the **OK** button.
8. When prompted, you can choose to save changes to all setups.
9. If you clicked Yes in step 9, the system prompts that x number of setups were changed. Press **Enter** or click the **OK** button.
10. Initialize the cutter to make your changes effective.

Single Step Diagnostic

Use this diagnostic to turn Single Step mode on or off. When Single Step mode is on, you can process cut file data one block at a time. This is useful when recovering from a FEP error.

Note: During Single Step mode, the cutter head rises if more than three seconds elapse without your initiating a command. This keeps material from fusing. The cutter head lowers when you initiate a command again.

To turn on the Single Step diagnostic

1. Do one of the following:
 - From the Diagnostics menu, select **Single Step**.
 - Right click in the gray area next to the gauges to display Settings Options. Click **Enable Single Stepping**.

A button displays on-screen, indicating that Single Step is ON.

2. When finished processing in this mode, select Single Step again or click on the Single Step button to turn off the diagnostic.

To process a cut file/marker using Single Step mode

1. Turn on Single Step mode.
2. Initialize the cutter.
3. Open a cut file/marker.
4. Set the marker origin.
5. Start processing the cut file/marker. The monitor displays the first line of data in the file and the cutter performs the specified command.
6. Press **Ctrl + S** to display the next line of data code and issue that command to the cutter.
7. Repeat step 6 until the end of the cut file/marker.

8. If the system finds an error, an error message displays. Use the line of data code and the error message to discover the error in processing the file.
9. When finished, press **Esc** to stop processing the cut file/marker in Single Step mode.

Continuous Run

Use this diagnostic to test the mechanical, electrical, and electronic functions of the GERBERcutter. When this feature is turned on, any cut file/marker that you process runs continuously until you press the Stop button. When the cutter reaches the end of the file, cutting starts again at the beginning.

To use Continuous Run

1. From the Diagnostics menu, select **Continuous Run**.
2. The Settings window indicates that Continuous Run is ON.
3. Open a marker and start processing.
4. To stop processing, you have to press the **Stop** button.
5. When finished using this mode, select Continuous Run again to turn off the diagnostic.

Display Bite Adjust Values

Use this diagnostic to display the Bite Adjustment value, which is calculated at the end of every bite. The default range is +/- 38.1 mm (+/- 1.5 in.) and can be changed by your service representative.

To run the diagnostic

1. From the Diagnostics menu, select **Display Bite Adjust Values**.
2. Process a bite. The Bite Adjustment value displays.
3. Note the value. Then press **Enter** or click the **OK** button.
4. When finished using this mode, select again to turn off the diagnostic.

Conveyor/CTOT Speed Calibrate

This diagnostic can be used on S-3200/5200/7200, GT 3250/5250/7250, and GTxL GERBERcutters to calibrate conveyor speed to Conveyorized Take-Off Table (CTOT) speed. Only use this diagnostic if told to by Field Service.

Note: To change speed settings, use Field Service level parameters in the Conveyor and CTOT category.

To run the diagnostic

1. From the Diagnostics menu, select **Conveyor/CTOT Speed Calibrate**.

A dialog box prompts that calibration has started.

2. To stop calibrating, press **Esc** or click the **OK** button.

Stepsize Setup

This diagnostic can be used to calculate new stepsize settings for the X and Y-axes and save these settings to system setup files.

To run the diagnostic

1. Use the Move Absolute command in the Cutter menu to move the X-axis a specific distance. Do this twice, marking the exact location of each move. Then measure the actual distance moved.
2. From the Diagnostics menu, select **Stepsize Setup**.

The Current stepsize shows in the dialog box.

3. Enter values for:
 - Desired move distance
 - Actual distance moved
4. Click the **Calculate** button.

5. Repeat steps 1 through 4 for the Y-axis.
6. When finished, click either the **Save** or **Save to All** button to save this setting to one or more setup files.

Note: The system must be re-initialized before the new setting will go into effect.

Edge Sensor Diagnostic

Use this diagnostic as part of the setup procedure for the Automatic Tube Centering option. For information on setting up and using this option, refer to document part number 83806000.

To run the diagnostic

1. Place a piece of underlay paper over the cutting table. Turn on the vacuum to hold the paper in place.
2. While the Edge Finders are retracted, hold a pencil against the inside edge of each of the two Edge Finders and mark two lines on the paper.
3. From the Diagnostic menu, click **Edge Sensor**. Click the **Test** button in the Edge Finder Diagnostic dialog box. This engages both edge finders, moving them the maximum allowed distance toward each other.
4. Mark the positions of the edge finders when engaged as explained in step 3. There should now be four lines on the paper.
5. Click **OK** to retract the Edge Finders and then click **Exit** to leave the diagnostic.

Position Info

Use this diagnostic to have the system calculate and display the change in X and Y distances from the current position to the current origin.

To run the diagnostic

1. From the Diagnostics menu, select **Position Info**. Or press **Ctrl + P**.
2. Note the Distance From Origin values reported.
3. Press **Esc** or click the **Close** button.

Move Head To

Use the Move Head To command in the Cutter menu to move the cutter head to an absolute (X,Y) position on the table. This command can be useful when performing diagnostics.

To move the head to a specific location

1. From the Cutter menu, select **Move Head To**.
2. In the Move Head To dialog box, type the X position to move to and then type the Y position to move to. Press **Enter** or click the **OK** button.

The new coordinates display in the System Status window.

Zone Valve Exerciser

The Zone Valve Exerciser diagnostic aids in testing, setting up, or troubleshooting hardware for the vacuum zone control.

To run this diagnostic

1. From the Diagnostics menu, select **Zone Valve Exerciser**.
2. In the dialog box, click to select whether you want to exercise all zones or only one zone.
3. If you selected all zones in step 2, go to step 4.

Or if you selected one zone in step 2, enter the zone number you want.

4. Click the **Start** button.

A check mark displays in the appropriate box as each zone is being exercised.

5. When finished, click the **Stop** button.

Cut Tests

Running Cut Tests

A Cut Test is actually just a marker designed to check a specific cutter operation. Run Cut Tests as you would any marker, replacing the spread with a piece of paper. Specific steps required for a Cut Test diagnostic and the expected results are explained in each topic. Also refer to the GERBERcutter operator's manual.

To run Cut Tests

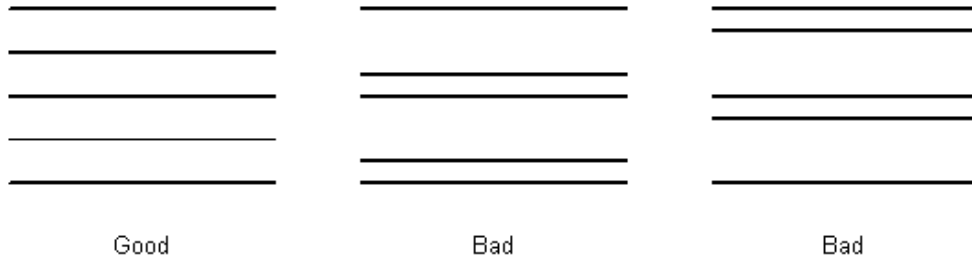
1. Start the C-200MT program.
2. Enable servo power.
3. Initialize the cutter.
4. Position paper and plastic on the table. See each diagnostic for the specific paper size to use.
5. Turn on the vacuum.
6. Slew the cutter head to the lower left corner of the paper.
7. On the beam control panel do **all** of the following:
 - Press the **ORIG** button.
 - Set **CUTTING SPEED** to **10**, unless otherwise noted.
 - Set **KNIFE** to **ON**.
8. From the Diagnostics menu, choose **Cut Tests**. Then select the appropriate diagnostic.
9. Press the **Start** button to run the diagnostic. The GERBERcutter displays the status screen while the diagnostic is running.
10. When processing is complete, at the beam control panel do **all** of the following:
 - Set **KNIFE** to **OFF**.
 - Press the **VAC ON/OFF** button.
11. Slew the cutter head out of the way, unless otherwise noted.
12. Check the results as explained in the topic for each diagnostic.

Multiple Horizontal Lines

This Cut Tests diagnostic was used on early model GERBERcutters to check the C-axis for proper mechanical alignment relative to the X-axis. During this diagnostic, the cutter cuts five horizontal lines in a piece of paper. The knife cuts three lines in a +X direction, and two in a -X direction. If all lines are evenly spaced, the knife is centered in the C-axis. If the lines are unevenly spaced, the knife is offset from the center of the C-axis.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper 0.5 m (20 in.) square or larger on the table.
 - Set Cutting speed to **5**.
2. To check results, measure the distance between the lines cut in the paper. The lines should be 12.7 mm (0.5 in.) apart. Call Field Service if the lines are unevenly spaced.

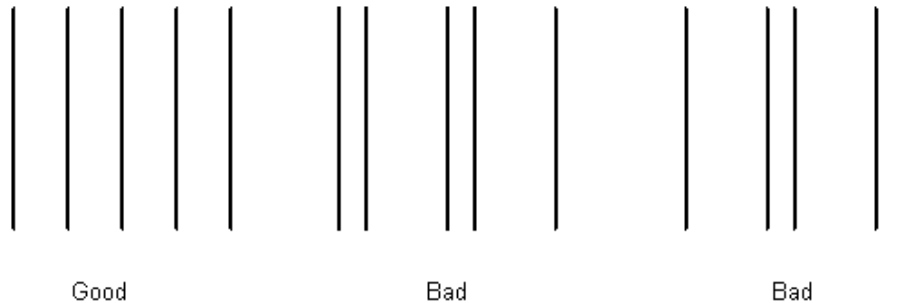


Multiple Vertical Lines

This Cut Tests diagnostic was used on early model GERBERcutters to check the C-axis for proper mechanical alignment along the Y-axis. During this diagnostic, the cutter cuts five vertical lines in a piece of paper. The knife cuts three lines in a +Y direction, and two in a -Y direction. If all lines are evenly spaced, the knife is centered in the C-axis. If the lines are unevenly spaced, the knife is offset from the center of the C-axis.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper 0.5 m (20 in.) square or larger on the table.
 - Set Cutting speed to 5.
2. To check results, measure the distance between the lines cut in the paper. The lines should be 12.7 mm (0.5 in.) apart. Call Field Service if the lines are unevenly spaced.

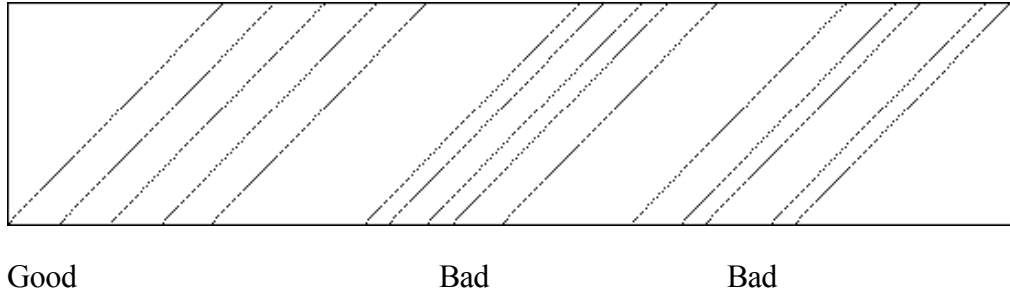


Multiple 45 degree Lines

This Cut Tests diagnostic was used on early model GERBERcutters to check the C-axis for proper mechanical alignment relative to the X and Y-axes. During this diagnostic, the cutter cuts five 45 degree lines in a piece of paper. If all lines are evenly spaced, the knife is centered in the C-axis. If the lines are unevenly spaced, the knife is offset from the center of the C-axis.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper 0.5 m (20 in.) square or larger on the table.
 - Set Cutting speed to 5.
2. To check results, measure the distance between the lines cut in the paper. The lines should be 12.7 mm (0.5 in.) apart. Call Field Service if the lines are unevenly spaced.



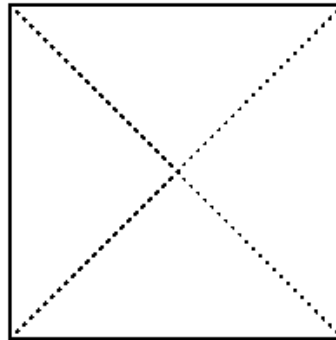
50" Square Test

This Cut Tests diagnostic can be used to check that the beam is square with the cutting table. The beam is square when neither end leads the other as the beam travels along the cutting table. In this diagnostic, the GERBERcutter cuts a 1.27 m (50 in.) square from a piece of paper. If corner-to-corner measurements are the same, the beam is square.

Note: The Beam Square Test is currently used instead of this diagnostic.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper 1.5 m (60 in.) square or larger on the table.
 - Before initializing the cutter, turn Run Time Biting OFF.
2. To check results, first measure a diagonal line from the upper left corner to the lower right corner. Then repeat from the upper right corner to the lower left corner to get the second measurement.



3. Compare the measurements with those in the table below.

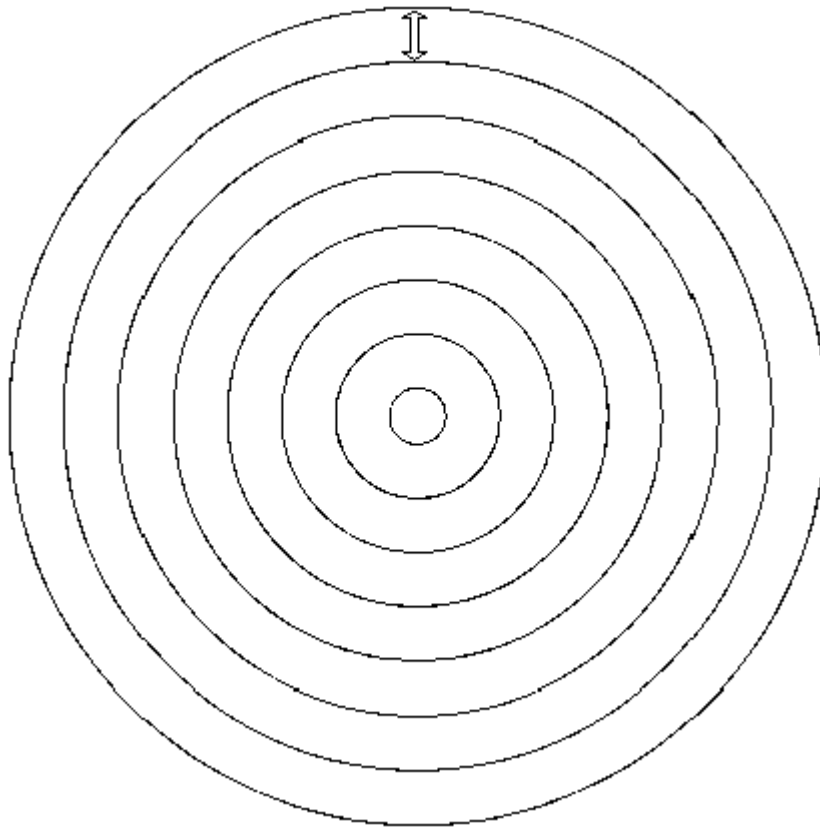
<u>Measurement</u>	<u>Indication</u>	<u>Action</u>
Equal to within 0.8 mm (0.03 in.)	The beam is square.	None
First is greater than second	The beam is not square. The +Y end of the beam leads the -Y end.	Use the difference between these measurements to adjust beam alignment. Refer to the GERBERcutter operator's manual.
Second is greater than first	The beam is not square. The -Y end of the beam leads the +Y end.	Same as above

Concentric Circles

This Cut Tests diagnostic can be used to check for proper alignment of the X, Y, and C-axes to ensure accurate cutting. In this diagnostic, the cutter cuts eight concentric circles from a piece of paper.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal, except place a piece of paper 0.5 m (20 in.) square or larger on the table.
2. To check results, make sure that each circle is perfectly round and exactly 2.54 cm (1 in.) larger than the one inside it. If so, the X, Y, and C-axes are properly aligned. Call Field Service if circles are not concentric.

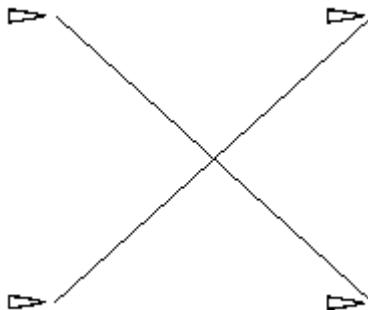


Beam Square Test

Use this Cut Tests diagnostic to check that the beam is square with the cutting table. In this diagnostic, the cutter plunges the knife at each corner of a square. If corner-to-corner measurements are the same, neither end leads as the beam travels along the table and the beam is square.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper at least 1.5 m (60 in.) square and preferably larger on the table. This diagnostic must be rerun after making any adjustments.
 - Before initializing the cutter, turn Run Time Biting OFF.
2. To check results, first measure a diagonal line from top right tip to bottom left tip of the knife plunges. Then repeat from top left tip to lower right tip to get the second measurement. Each line should be 160 cm (63 in.).



3. Compare the measurements with those in the following table.

<u>Measurement</u>	<u>Indication</u>	<u>Action</u>
Equal to within 1.0 mm (0.03 in.)	The beam is square.	None
First is greater than second	The beam is not square. The +Y end of the beam leads the -Y end.	Use the difference between these measurements to adjust beam alignment. Refer to the GERBERcutter operator's manual.
Second is greater than first	The beam is not square. The -Y end of the beam leads the +Y end.	Same as above.

Heel to Heel Test

Use this Cut Tests diagnostic to check the mechanical alignment of the knife in the C-axis. The knife must be properly aligned for accurate cutting. In this diagnostic, the system plunges the knife twice so that alignment can be checked.

To run the diagnostic

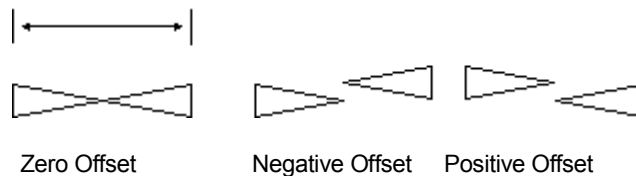
1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper 0.5 m (20 in.) square or larger on the table.
 - After initializing the cutter set Knife speed to:
 - **10,500 rpm** for the S-3200 and GT 3250 only
 - **2,500 rpm** for all other cutters

CAUTION: Knife speed must be set to one of the above values to simulate the knife load caused by cutting a spread and ensure that results of this diagnostic will be accurate.

- Set Cutting speed to **1**.
2. To check results, compare your test knife cuts with the graphic below.
 3. If correct, go to step 5.
 4. If the knife is out of alignment, your trained GERBERcutter maintenance person can adjust knife alignment using the Gerber Knife Adjustment box. Otherwise, call Field Service.

Note: A slight negative offset is beneficial under certain cutting applications. Contact Field Service for recommendations and limitations.

Heel-to-heel Measurement



5. When alignment is correct, measure the heel-to-heel distance. If this measurement falls outside the proper range, call Field Service. See the table below.

<u>GERBERcutter*</u>	<u>Range for High Ply Knife</u>	<u>Range for Low Ply Knife</u>
S-3200, GT 3250	11.0 to 11.5 mm (0.43 to 0.45 in.)	4.6 to 5.1 mm (0.18 to 0.20 in.)
S-5200, GT 5250	Fixed at 12.7 mm +/- 0.80 (0.5 in. +/- 0.03)	N/A
S-7200, GT 7250	12.7 to 14.3 mm (0.5 to 0.56 in.)	N/A
S-91 ZVC Static S-91 S-91 TCW		

* Not applicable for GTxL GERBERcutter

Yaw in X Test

Use this Cut Tests diagnostic to check the knife for yaw in the X-axis. X-axis yaw is the angle of the knife relative to the X-axis. For most applications, there should be no yaw; the knife should be parallel to the X-axis. In this diagnostic, the cutter head travels in a straight line parallel to the X-axis. As it travels, the cutter head plunges the knife four times. The angle of knife plunges relative to the line of travel indicates the current yaw value.

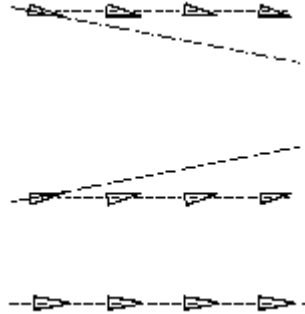
To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper 0.5 m (20 in.) square or larger on the table.
 - After initializing the cutter, set Knife speed to:
 - **10,500** rpm for the S-3200 and GT 3250 only
 - **2,500** rpm for all other cutters

CAUTION: Knife speed must be set to one of the above values to simulate the knife load caused by cutting a spread and ensure that results of this diagnostic will be accurate.

- Set Cutting speed to **1**.

2. To check results, draw a line through the first cut from the center of the heel through the center of the tip. Extend the line past the last cut. The extended line must pass through all additional cuts in order to have perfect alignment (shown in the bottom example).



3. If extended lines do not pass through all cuts, the knife is out of alignment in X. Enter a new value in the C Home to System 0 Degrees parameter to correct the condition.

Note: Increasing this value moves toward a positive yaw angle; decreasing the value moves toward a negative yaw angle.

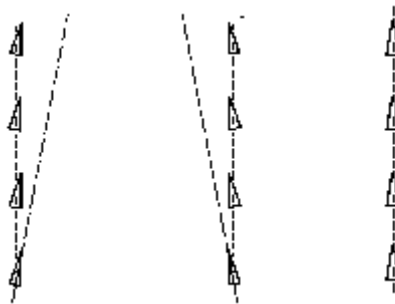
4. Repeat the diagnostic to check results of your changes. When correct, save changes to your setup files.

Yaw in Y Test

Use this Cut Tests diagnostic to check the knife for yaw in the Y-axis. Y-axis yaw is the angle of the knife relative to the Y-axis. For most applications, there should be no yaw; the knife should be parallel to the Y-axis. In this diagnostic, the cutter head travels in a straight line parallel to the Y-axis. As it travels, the cutter head plunges the knife four times. The angle of knife plunges relative to the line of travel indicates the current yaw value.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Place a piece of paper 0.5 m (20 in.) square or larger on the table.
 - After initializing the cutter, set Knife speed to:
 - **10,500 rpm** for the S-3200 and GT 3250 only
 - **2,500 rpm** for all other cutters
 - CAUTION:** Knife speed must be set to one of the above values to simulate the knife load caused by cutting a spread and ensure that results of this diagnostic will be accurate.
 - Set Cutting speed to **1**.
2. To check results, draw a line through the first cut from the center of the heel through the center of the tip. Extend the line past the last cut. The extended line must pass through all additional cuts in order to have perfect alignment (example at right). If extended lines do not pass through all cuts, go to step 3.



3. If the knife is out of alignment in Y, enter a new value in the C Home to System 0 Degrees parameter to correct the condition.

Note: Increasing this value moves toward a positive yaw angle; decreasing the value moves toward a negative yaw angle.

4. Repeat the diagnostic to check results of your changes. When correct, save changes to your setup files.

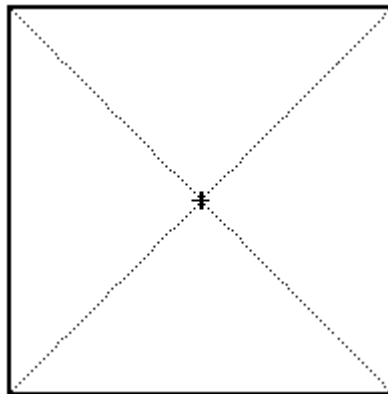
10" Square, Center Drill Hole

This diagnostic can be used to check that the current drill offset value allows the GERBERcutter to place drill holes accurately. Drill offset is the location of the drill bit relative to the origin light. The system cuts a 25.4 cm (10 in.) square and then places a drill hole in the square. If the hole is in the exact center of the square, the system is using the proper value for drill offset.

Note: **Note:** Drill 1 Offset Setup and Drill 2 Offset Setup are currently used instead of this diagnostic.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal, except place a piece of paper that is 0.5 m (about 20 in.) square or larger on the table.
2. To check results, draw diagonal lines from opposite corners of the square. The lines should cross over the drill hole.
3. If the lines do not cross, the drill offset is incorrect. Call Field Service to correct the condition.

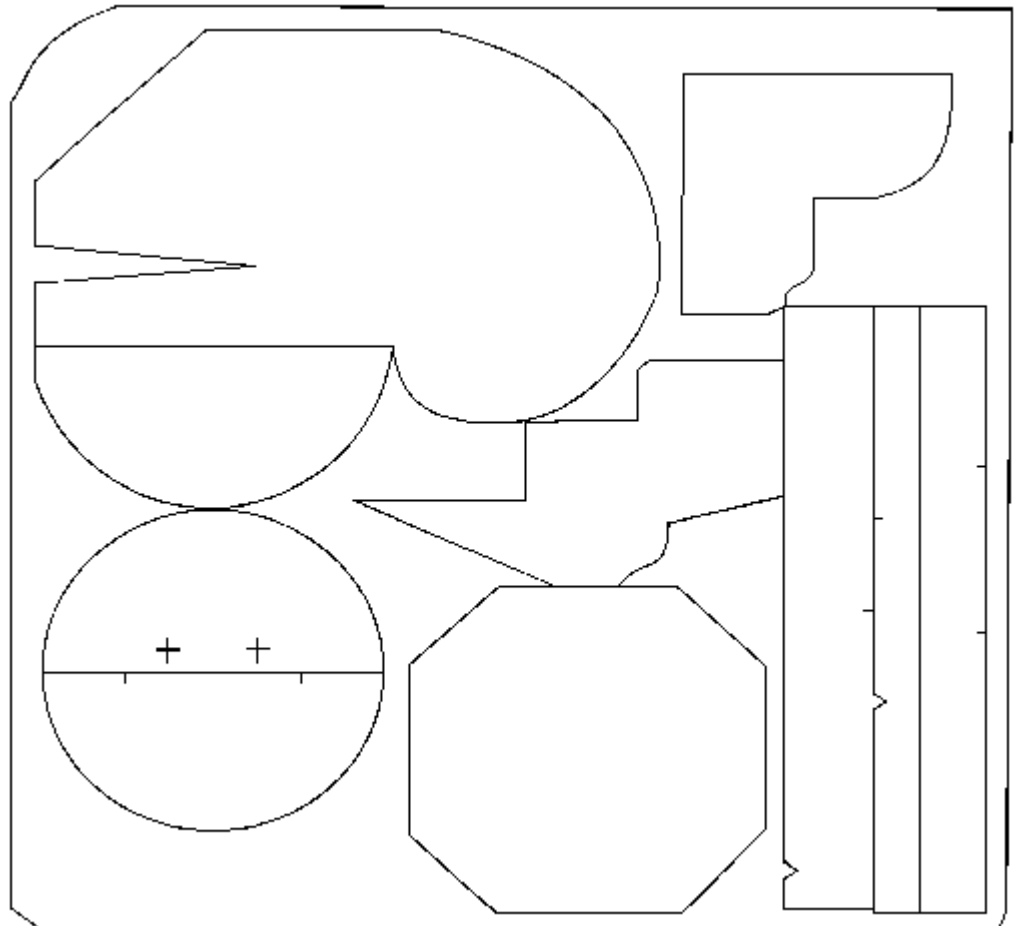


Geometric Shapes

Use this Cut Tests diagnostic to test for proper functioning of GERBERcutter components with the C-200MT program. In this diagnostic, the cutter processes a cut file consisting of several different geometric shapes. The system is functioning correctly if it cuts all the pieces without any problems.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal, except pull paper from a spreading table onto the cutting table. Make sure you cover the entire table.
2. To check results, compare the cut pieces with the pattern below. Call Field Service if the results are not as expected.



5' Bite Test

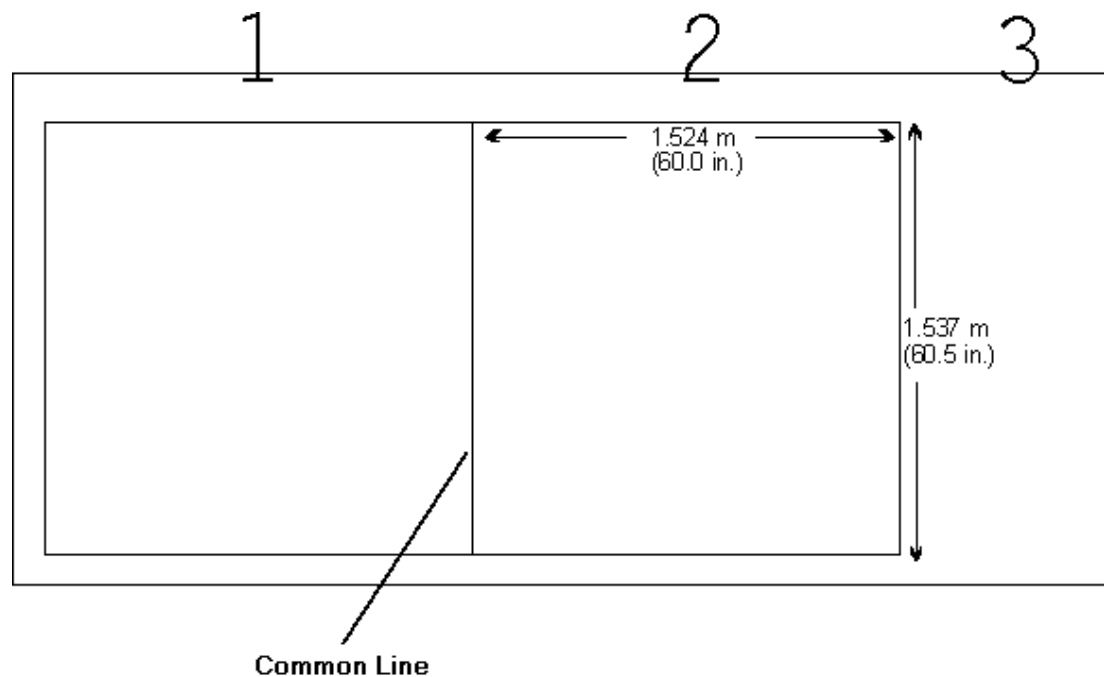
Use this Cut Tests diagnostic to test the conveyor for proper bite feeding by processing two squares (5 ft or 1.524 m) in three bites. If the conveyor is not set up correctly, the common line shared by the squares does not cut properly.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except:
 - Make sure Run Time Biting is OFF.
 - Pull paper from a spreading table onto the cutting table. Make sure you cover the entire table.

The system cuts square 1, advances 1.524 m (5 ft) to bite 2, cuts square 2, advances 1.524 m (5 ft) to bite 3, and stops cutting.

2. To check results, compare with the pattern below. The conveyor is properly adjusted if the common line between square 1 and square 2 has been cut only once. Call Field Service if results are not as expected.



Knife Offset Setup

Use this Cut Tests diagnostic to check that the cutter is using the correct value for knife offset. Knife offset is the location of the knife blade relative to the origin light. If the system is using the wrong value for knife offset, cuts will not be made in the correct location.

Note: This diagnostic should be used whenever the laser light or head is changed.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except, place a piece of paper 1.5 m (60 in.) square or larger on the table.

The cutter positions the knife at, or near, the origin and executes a knife down cycle. After it stops cutting, the Systems Message box prompts that processing is complete.

2. Use the Move to Origin command.
3. To check results, make sure that the origin light shines exactly on the intersection of the horizontal and vertical lines. If it does not, do **one** of the following:
 - Continue with steps 4 through 6.
 - Call Field Service.
4. Use the Position Info diagnostic to determine the exact location.
5. Make appropriate changes to the Offset in X - Knife and Offset in Y - Knife parameters in the System Setup category.
6. Reorigin and repeat this diagnostic to check your changes.

Drill 1 Offset Setup

Use this Cut Tests diagnostic to check that the cutter is using the correct value for the main drill offset. Drill offset is the location of the drill bit relative to the origin light. If the system is using the wrong value for drill offset, it can not cut accurately.

Note: This diagnostic should be used whenever the laser light or head is changed.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except place a piece of paper 1.5 m (60 in.) square or larger on the table.

The cutter drills a single drill hole. The Systems Message box prompts that processing is complete.

2. Use the Move to Origin command.
3. To check results, make sure that the origin light shines exactly onto the drill hole. If it does not, do **one** of the following:
 - Continue with steps 4 through 6.
 - Call Field Service.
4. Use the Position Info diagnostic to determine the exact location.
5. Make appropriate changes to the Offset in X - Drill 1 and Offset in Y - Drill 1 parameters in the System Setup category.
6. Reorigin and repeat this diagnostic to check your changes.

Drill 2 Offset Setup

Use this Cut Tests diagnostic to check that the cutter is using the correct value for the auxiliary drill offset. Drill offset is the location of the drill bit relative to the origin light. If the system is using the wrong value for drill offset, it can not cut accurately.

Note: This diagnostic should be used whenever the laser light or head is changed.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except place a piece of paper 1.5 m (60 in.) square or larger on the table.

The cutter drills a single drill hole. The Systems Message box prompts that processing is complete.

2. Use the Move to Origin command.
3. To check results, make sure that the origin light shines exactly onto the drill hole. If it does not, do **one** of the following:
 - Continue with steps 4 through 6.
 - Call Field Service.
4. Use the Position Info diagnostic to determine the exact location.
5. Make appropriate changes to the Offset in X - Drill 2 and Offset in Y - Drill 2 parameters in the System Setup category.
6. Reorigin and repeat this diagnostic to check your changes.

Pen Offset Setup

This Cut Tests diagnostic can be used to check that the GERBERcutter is using the correct value for pen offset. Pen offset is the location of the pen relative to the origin light.

To run the diagnostic

1. Run the Cut Tests diagnostic as normal except place a piece of paper 1.5 m (60 in.) square or larger on the table.

A vertical and a horizontal line are drawn in an L shape. The Systems Message box prompts that processing is complete.
2. Use the Move to Origin command.
3. To check results, make sure that the origin light shines exactly on the intersection of the horizontal and vertical lines. If it does not, do **one** of the following:
 - Continue with steps 4 through 6.
 - Call Field Service.
4. Use the Position Info diagnostic to determine the exact location.
5. Make appropriate changes to the Offset in X - Pen and Offset in Y - Pen parameters in the System Setup category.
6. Reorigin and repeat this diagnostic to check your changes.

Beam Square Diagnostics

Beam Squareness

This diagnostic is used along with Beam Square Test to square the beam for cutters, such as GTxL, with two, independent X motors. Use this diagnostic to have the system set a Beam Square Adjustment value based on x1 and x2 index distances. This value needs to be reset whenever:

- Mechanical part(s) affecting x1 and x2 index distances have been replaced or modified
- Beam has been mechanically squared
- FEP software has been loaded

Once this diagnostic has been run, the beam is automatically squared when you initialize the cutter, including after EStop or error conditions.

Parameters for Beam Squareness Diagnostic

Two Field Service level parameters in the System Setup Parameters category are used along with the Beam Squareness diagnostic: Beam Square Adjustment and Beam Squared.

To set Beam Square Adjustment value

1. Mechanically square the beam as explained in the Maintenance manual for the GERBERcutter.
2. Make sure that the Beam Squared parameter in the System Setup Parameters category is set to NO.
3. Initialize the cutter.

The cutter performs normal Home routine, then the beam moves one inch and back to initial position.

4. Square the beam as explained on next page.

To square beam after value is set

1. Position paper and plastic on the cutter table.
2. From the Diagnostics menu, choose Cut Test and then select **Beam Square Test**.

The diagnostic file (diag9.int) is opened.

3. From the Cutter menu, select **Move Head To**. In the dialog box enter X5.0, Y5.0. Press **Enter** or click **OK**.
4. On the beam control panel do **all** of the following:
 - Press the **ORIG** button.
 - Turn the **VACUUM** on.
 - Set **KNIFE** to on.
5. Cut the marker.

The knife plunges at each corner of the square.

6. Measure diagonals (about 160 cm).
7. From the Diagnostics menu, select **Beam Squareness**.
8. In the dialog box, enter the exact measurements for Diagonal 1 and Diagonal 2.

The system modifies the Beam Square Adjustment parameter and sets the Beam Squared parameter to YES. Click **OK**.

To automatically square beam

1. Make sure the Beam Squareness diagnostic has been run at least once.
2. Initialize the cutter. During initialization, the x2 position is adjusted (click) to the new Beam Square Adjustment value.

Path Intelligence Diagnostics

Use diagnostics in the Path Intelligence submenu to:

- Calibrate PI
- Optimize PI

Calibration

Use this Path Intelligence diagnostic to set up Path Intelligence (PI) for your system. Also use this diagnostic to test the current draw of your cutter head and beam. You should use this diagnostic:

- Daily when using PI
- For each new spread

This diagnostic checks the amount of electrical current required to move the cutting head with the knife up in a square pattern. It starts at feed rate 2, incrementing one feed rate as it moves through each square up to feed rate 15. After it has finished testing current draw, data is stored for use by Path Intelligence.

To run the diagnostic

1. From the Diagnostics menu, select Path Intelligence.
2. Select **Calibration**.

The system moves the cutter head and beam, in a square pattern 14 times. After it stops, respond to the system message as explained in the following table.

<u>System Message</u>	<u>Appropriate Action</u>
PASS - X and Y currents within normal ranges	Begin cutting.
WARNING - X or Y current increased since last test	Call Field Service.
WARNING - X or Y current higher than normal	Call Field Service.
FAIL - X or Y current unacceptably high	Call Field Service.

Watch Forces

Use this diagnostic to optimize Path Intelligence (PI). To do this, first measure the force needed to cut different materials at different feed rates and ply heights. Then for each material type, use this data along with the PI: Optimum Force parameter to determine the appropriate maximum speed setting and the maximum number of plies to cut.

To run the diagnostic

1. From the Diagnostics menu, select Path Intelligence.
2. Then select **Watch Forces**.

The following parameter setting displays in the Settings window:

Force:x.xxx (z.zzz)

where x.xxx is a real- time measure of the force the cutter is using and z.zzz is the maximum force that has been measured since you turned on the Watch Forces diagnostic.

Note: To reset the maximum measured force to zero, you must turn off this diagnostic and turn it back on.

3. When finished optimizing PI, select **Watch Forces** again to turn off the diagnostic.

Software Diagnostics

Diagnostics in this menu are used during software development and may be used by Field Service when troubleshooting.

Restorable Info

This diagnostic provides information that can be used to restore values after a system shut down. The types of information include:

- Knife Wear
- Origin
- Setup
- Marker Info
- Queue Info

To run the diagnostic:

1. From the Diagnostics menu, choose Software Diagnostics and then select **Restorable Info**.
2. View information about the current state.
3. To close the window, click the button marked by an X at the top right corner.

Debug Window Diagnostic

Use this diagnostic to open a window to display internal diagnostic and communication information. Event Logging in the Software Parameters category must be enabled.

To run the diagnostic:

1. From the Diagnostics menu, choose Software Diagnostics and then select **Debug Window**.
2. To close the window after viewing debug data, repeat step 1.

Servo Data Logging Diagnostic

Use this diagnostic to copy the most recent communication data between motion control board and operator console to hard disk or to a 3.5 inch disk for analysis.

To run the diagnostic:

1. From the Diagnostics menu, choose Software Diagnostics and then select **Servo Data Logging**.
2. To copy to a 3.5 inch disk, type a:\datalog.dat in the File Name field.
3. To change the sample interval, click the up or down arrow buttons and then click the **Set Mask** button.
4. Click the Upload Data button.
5. To close the window, click the button marked with an X at the top right corner.

Error Simulation Diagnostic

Use this diagnostic to simulate:

- Actual error message and evaluate cutter error handling
- Dr. Watson error for Windows NT

To run the diagnostic:

1. From the Diagnostics menu, choose Software Diagnostics and then select **Error Simulation**.
2. To evaluate cutter error handling, type in the error number you want to simulate and click **OK**.

Note: Errors 4000 and 4001 occur when the system identifies a situation and then calls a shut down. Errors 4002 and 4003 are invoked without any prior warning to the system. When simulating errors 4002 or 4003, a marker must be opened and run. The error will occur at piece #2 in the marker.

3. To evaluate a Dr. Watson error, click the Dr Watson Error Simulation button. When the warning displays, click OK to continue or Cancel to return to the diagnostic. Click OK to save the unhandled exception to the c:\bin\c200mt.rpt file.

Communications Statistics

Use this diagnostic to view how many attempts were made to establish:

- Communication between the controller and the operator console.
- Why communications failed.

To run the diagnostic:

1. From the Diagnostics menu, choose Software Diagnostics and then select **Communications Statistics**.
2. To close the window, click the button marked by an x at the top right corner.

Dual Beam Diagnostics

GERBERcutters that have the dual beam option have additional diagnostics to check the secondary beam. Refer to the online help for your GERBERcutter.

Diagnostic Addendum

Dynamic Offset / Yaw Cut Test

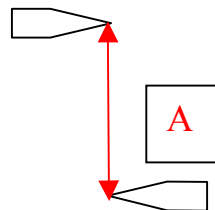
Offset

Use this Cut Tests diagnostic as follows to check offset whenever maintenance is performed on the sharpener or roller guide or if a cut quality or sizing condition exists.

1. From the Diagnostics menu, choose Cut Tests and select **Dynamic Yaw / Offset**.
2. Select **1 inch** to open the YAW 1in. cut file or select **3 cm** to run the YAW 3cm cut file.
3. Place a 4 inch x 8 inch stack of a typical material being cut on the cutting surface. Use underlay paper and overlay plastic.

Note: Best results are obtained with medium to high ply test cuts of harder materials.

4. Select the appropriate knife speed and feed rate for material to be cut. Run the cut test.
5. Measure the distance between the points of the two knife plunges (distance A). The measurement should be 1 inch or 3 cm, depending on the specific cut test run.



6. If the measurement is correct, go to step 9. Or to check Yaw also, go to step 6 in the Yaw procedure.
7. If the measurement is incorrect, refer to presser foot roller guide adjustments for adjustment.
8. After adjustment, repeat steps 2 through 7 until it is corrected.

9. Exit the diagnostic.

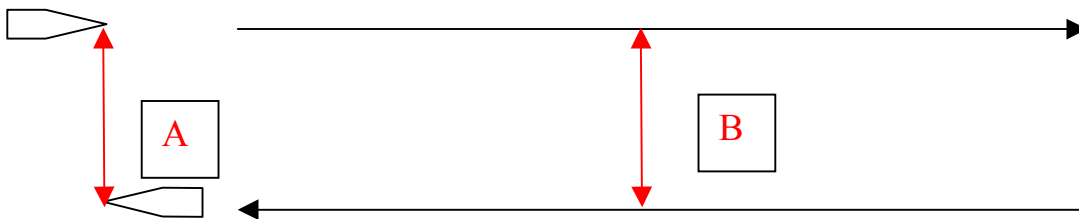
Yaw

Use this diagnostic as follows to check yaw whenever maintenance is performed on C-axis components or the sharpener.

1. From the Diagnostics menu, choose Cut Tests and select **Dynamic Yaw / Offset**.
2. Select 1 inch to open the YAW 1in. cut file or select 3 cm to run the YAW 3cm cut file.
3. Place a 4 inch x 8 inch stack of a typical material being cut on the cutting surface. Use underlay paper and overlay plastic.

Note: Best results are obtained with medium to high ply test cuts of harder materials.

4. Select the appropriate knife speed and feed rate for material to be cut. Run the cut test.
5. Measure the distance between the points of the two knife plunges (distance A). The measurement should be 1 inch or 3 cm based on the specific cut test run.
6. Measure the distance between the two cut lines (distance B) midway from the ends of the lines. The measurement should be equal to the “offset” measurement. Any CW Piece deviation (refer to graphs on next page) is described as $B - A$.



7. If the measurements are correct, go to step 10.
8. If the measurement is incorrect, perform the C Home to System Zero diagnostic using the value determined from graph 1 (in inches) or graph 2 (in centimeters) to modify the C Home to System 0 Degrees parameter. See graphs on next page.
9. Repeat steps 2 through 8 until the measurements are correct.

10. Exit the diagnostic.

